nection PCBs, via size (diameter) is getting reduced. As per IPC, any via hole having drill diameter of less than or equal to 0.15mm is a micro via. Such via holes are laser drilled.

Via-in pad is generally used where it becomes difficult to put a via hole in between the pads/land area due to limitation of available space. In this case, a via hole is drilled inside a land area, and we can say that the land area is converted into a pad. Such via-hole's diameter must be small (as per manufacturer limitations) as there may a chance of solder theft at the time of reflow soldering. Components like bill grid array (BGA) require via-in pads. (Land area is a copper area without a hole, where an SMD component is soldered. A pad is a copper area with a hole in the centre.)

Via for stitching is also used to interconnect large copper areas on different layers. It helps reduce the return path of signals and thereby maintain low impedance. A via shielding is created by placing one or more rows of via holes along the signal route path, which helps in reducing electromagnetic interference (EMI) and crosstalk.

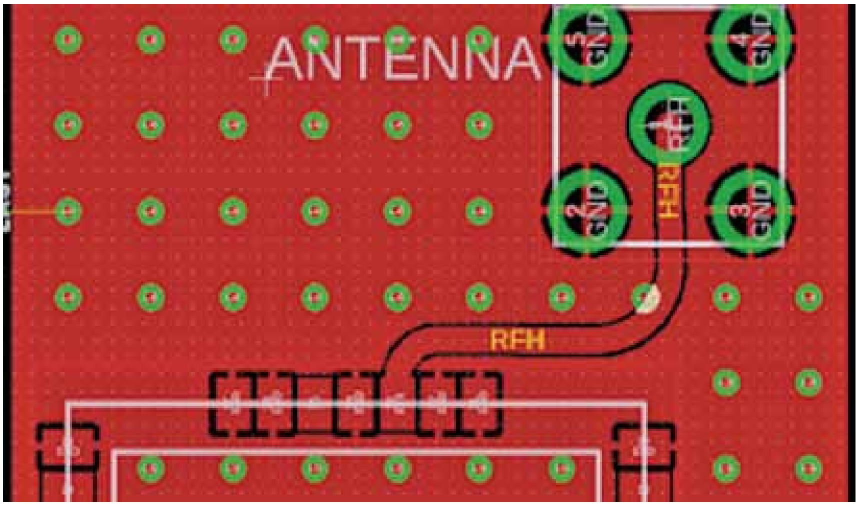

Fig. 3: Example of via for stitching

Thermal via is used to improve the heat dissipation of surface-mount components rather than transferring a signal. Structurally, a penetrating hole is opened in the board. Thermal resistance depends on the area and thickness of the copper foil on the PCB that serves to dissipate heat, as well as on the thickness and material of the board.

In order to effectively use thermal vias, it is important that the thermal vias be located close to heating elements, such as directly below the components. As indicated in Fig. 4, the effect of thermal equilibrium is utilised, and so it should be apparent that connecting areas with large temperature differences is a good plan.

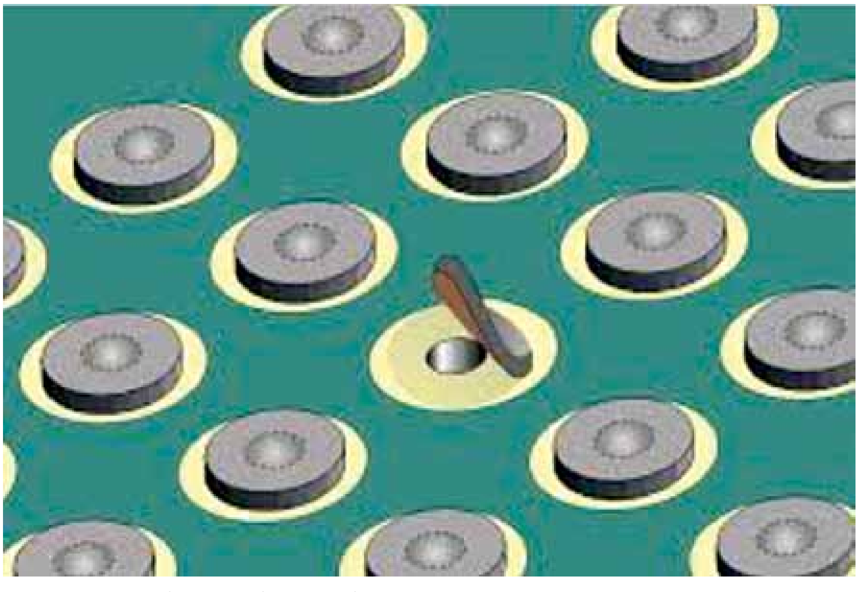

Fig. 2: Illustration of solder theft in case of a via-in pad

Via holes and signal integrity

In case of low frequency, vias do not affect signal transmission. As frequency rises (above 1GHz) and signal rising edge becomes steep (1ns at most), however, vias cannot be simply regarded as a function of electrical connection, and influence of vias on signal integrity has also to be considered. Vias behave as breakpoints with discontinuous impedance on transmission line, causing signal reflections.

Nevertheless, issues brought by vias concentrate more on parasitic capacitance and parasitic inductance. The influence of via parasitic capacitance on a circuit is primarily to prolong the rising time of signals and reduce the running speed of circuit. Parasitic inductance, however, can weaken the contribution of bypass circuit and decrease filtering function of the whole power system.

The larger the via diameter is, the more discontinuity via will cause in high-frequency and high-speed PCB designs. It is tentative